
COMPSCI 602

Project Report 2

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Project Title: Identifying the Internal Causal Mechanism of Shallow Safety Alignment Decay in Large Language Models

1 System, task, environment, and implementation

System. The system under study is **Llama-3.1-8B-Instruct** [4], a safety-aligned large language model with 8 billion parameters. This model was selected because it is a widely studied, publicly available instruction-tuned model accessible via HuggingFace, and it exhibits the shallow safety alignment phenomenon documented by Qi et al. [6]. The model’s intermediate size makes it tractable for detailed mechanistic analysis while being large enough to display complex safety-relevant behaviors. If compute becomes a bottleneck, **Llama-3.2-3B-Instruct** serves as a viable fallback.

Task. The task is **safety-constrained response generation**: given a harmful query (e.g., requesting instructions for dangerous activities), the model should produce a refusal. The key behavior we study is the transition from refusal to compliance as the model generates tokens beyond the initial refusal prefix. Unlike standard safety evaluation which examines only whether a model refuses, we study how the model’s internal safety representation evolves *across token positions during autoregressive generation*.

Environment. The environment consists of adversarial prompt sets that reliably trigger shallow alignment failures. We use two specific attack types: (1) **prefilling attacks**, where the model is forced to begin generation with a compliant prefix [6], and (2) **adversarial suffix attacks** that shift the model’s behavior after an initial refusal [10]. We vary the attack type, the number of forced prefix tokens, and the layer at which interventions are applied. Benign prompts from Alpaca [7] serve as controls. For evaluation of harmful content, we use prompts drawn from **AdvBench** [10] and **JailbreakBench** [2], which are standard benchmarks containing hundreds of harmful behavior requests with established evaluation protocols.

Implementation. All experiments are conducted at inference time using causal interventions, no model training or fine-tuning is required. The primary experimental framework is the **pyvene** library (v1.0) [8], a mature, general-purpose framework for activation interventions on PyTorch models that supports interchange interventions, activation patching, and Distributed Alignment Search (DAS). The code for shallow alignment attacks is publicly available [5]. Llama-3.1-8B-Instruct requires approximately 16 GB of GPU memory for inference, which is available on a single A100 GPU via the UMass GPU cluster. For analysis of the refusal direction, we also leverage the methodology introduced by Arditi et al. [1] using difference-in-means extraction of contrastive activation directions.

2 Phenomena

The primary phenomenon under study is the **positional decay of safety-critical internal representations during autoregressive generation** in safety-aligned LLMs.

Recent work has established that safety alignment in current LLMs is “shallow”: safety constraints are concentrated in the first few generated tokens and rapidly fade over the course of generation [6]. When subjected to prefilling attacks, where the model is forced to begin generation with a compliant prefix, models that initially refuse harmful queries gradually comply as generation proceeds. Qi et al. [6] demonstrated that this pattern unifies multiple known vulnerabilities, including susceptibility to adversarial suffix attacks, prefilling attacks, decoding parameter attacks, and fine-tuning attacks. However, their characterization is purely *behavioral*: they measure output token distributions but do not explain *why* the safety signal decays internally.

Separately, Chen et al. [3] identified a sparse set of “safety neurons” (approximately 5% of all neurons) whose activations are causally linked to refusal behavior, and demonstrated that patching these neurons’ activations can restore over 90% of safety performance. Arditì et al. [1] further showed that refusal is mediated by a single linear direction in the model’s residual stream, and that adversarial suffixes suppress propagation of this direction by hijacking the attention of safety-critical heads. Zhou et al. [9] extended this by identifying specific “safety heads” in the attention mechanism whose ablation alone allows $16\times$ more harmful responses.

Despite these advances, the **causal mechanism by which the influence of these safety-critical components decays across token positions during generation** remains entirely unexplained. This is the central phenomenon we aim to investigate: not merely *that* safety alignment is shallow, but *why* it is shallow at the level of internal causal mechanisms.

A secondary phenomenon is the **layer-specificity of the safety signal**. The refusal direction identified by Arditì et al. [1] is extracted from specific layers (typically mid-to-late layers), and safety heads identified by Zhou et al. [9] are concentrated in particular layers. Whether this layer distribution shifts across token positions during generation is unknown.

3 Variables

We identify the following variables that characterize the system, task, and environment, and whose values could affect the phenomena under study:

System variables:

- **Layer index** ($l \in \{1, \dots, L\}$): Which transformer layer the intervention or measurement is applied at. Prior work suggests safety-relevant information is concentrated in mid-to-late layers [1], but this has not been studied across token positions.
- **Component type**: Whether the safety signal is carried by attention heads, MLP sublayers, or specific neurons. Chen et al. [3] focus on neurons ($\sim 5\%$), Zhou et al. [9] on attention heads ($\sim 0.006\%$ of parameters), and Arditì et al. [1] on a linear direction in the residual stream.
- **Model scale**: Llama-3.1-8B-Instruct as the primary system, with Llama-3.2-3B-Instruct as a fallback. Model scale may affect the depth and distribution of safety representations.

Task and environment variables:

- **Token position** ($t \in \{1, 2, \dots, T\}$): The position within the generated sequence at which internal representations are measured or intervened upon. This is the most critical independent variable, we hypothesize that the safety signal’s causal contribution decays as t increases.
- **Attack type**: Prefilling attacks (forced compliant prefix of varying lengths) versus adversarial suffix attacks. Different attack types may cause different patterns of safety signal suppression.
- **Number of forced prefix tokens** (k): In prefilling attacks, the number of compliant tokens forced before the model generates freely. Qi et al. [6] showed that compliance increases with k , but the internal mechanism has not been traced.
- **Prompt harmfulness category**: Different types of harmful requests (e.g., violence, misinformation, illegal activities) drawn from AdvBench [10] and JailbreakBench [2]. Category-specific differences in the safety mechanism’s decay would be informative.

- **Intervention subspace rank:** When performing DAS or activation patching, the dimensionality of the subspace intervened upon (from rank-1 for the refusal direction to higher-rank subspaces).

Dependent variables (measured outcomes):

- **Cosine similarity with refusal direction:** Following Ardit et al. [1], measured at each layer and token position to track how the refusal signal propagates.
- **Causal effect of activation patching:** The change in model output (refusal vs. compliance) when safety-neuron activations from early token positions are patched into later positions via pyvene [8].
- **Refusal score and safety score:** Binary classification of model outputs as refusal/non-refusal (via string matching) and safe/unsafe (via Llama Guard), following the evaluation protocol in [1].

4 Frontier

The study of internal safety mechanisms in LLMs has advanced rapidly, converging on a picture where safety alignment is encoded in specific, identifiable components, but the temporal dynamics of this encoding during generation remain a critical gap.

Safety alignment is shallow and token-position-dependent. Qi et al. [6] provided the foundational behavioral evidence that safety-aligned LLMs concentrate their safety constraints in the first few generated tokens. Their experiments across multiple attack modalities demonstrated that models transition from refusal to compliance as generation proceeds, particularly under prefilling attacks. They further showed that data augmentation with “safety recovery examples” (where the model transitions back to refusal mid-generation) can deepen alignment, suggesting that the shallow pattern is an artifact of training rather than an architectural constraint. However, this work is entirely behavioral; it characterizes *what* happens in the output token distribution without explaining *why* the safety signal decays internally.

Safety is mediated by sparse, identifiable components. Three independent lines of work have identified the internal substrates of safety behavior with increasing precision. Chen et al. [3] identified approximately 5% of neurons as “safety neurons” using inference-time activation contrasting, showing that patching their activations restores over 90% of safety performance across red-teaming benchmarks. Ardit et al. [1] showed that refusal across 13 chat models (up to 72B parameters) is mediated by a single linear direction in the residual stream, identifiable via difference-in-means between harmful and harmless prompt activations. Ablating this direction eliminates refusal; adding it induces refusal on harmless inputs. Zhou et al. [9] demonstrated that individual attention heads are critical for safety, with ablation of a single “safety head” allowing $16\times$ more harmful responses while modifying only 0.006% of parameters. These results collectively establish that safety behavior depends on a surprisingly small, concentrated set of components, making it both identifiable and vulnerable.

The refusal direction is suppressed by attacks, but the suppression mechanism across token positions is unknown. Ardit et al. [1] showed that adversarial suffixes suppress the refusal direction by hijacking the attention of safety-critical heads, redirecting attention from instruction tokens to suffix tokens. However, this analysis was conducted at a *single token position* (the last token before generation) and for a *single model and suffix*. Whether and how the refusal direction’s magnitude and causal contribution change across generated token positions, the central question for explaining shallow alignment, has not been investigated. Similarly, Zhou et al. [9] attributed safety to specific attention heads but did not study how their contributions vary during the autoregressive generation process.

Gaps and untested assumptions. Several critical gaps remain at the frontier:

1. **No causal account of positional decay.** The decay of safety signals across token positions has been demonstrated behaviorally [6] but has no mechanistic explanation. It is unknown whether safety neurons [3] lose their activation magnitude, whether the refusal direction [1] rotates or diminishes in the residual stream, or whether safety heads [9] shift their attention patterns as generation proceeds.

2. **Untested restoration hypothesis.** If the safety signal decays because specific activations diminish at later token positions, can patching early-position safety activations into later positions causally re-trigger refusal? This would directly test whether the decay is due to representational drift versus active suppression by other model components.
3. **Interaction between attack type and decay mechanism.** Prefilling attacks and adversarial suffix attacks may cause safety signal decay through fundamentally different internal pathways. Prefilling forces compliant text into the context, potentially overwriting safety representations in the residual stream, while adversarial suffixes hijack attention [1]. Whether these attacks converge on the same internal decay pattern or operate through distinct mechanisms is untested.
4. **Layer-position interaction.** The refusal direction is typically strongest at mid-to-late layers [1], and safety heads are concentrated in specific layers [9]. Whether the layer distribution of the safety signal changes across token positions, e.g., whether safety information persists longer in earlier layers, has not been explored.

This project aims to address these gaps by performing causal mediation analysis across *both* layers and token positions using interchange interventions via pyvene [8]. Specifically, we will trace how the residual stream representation of “refuse vs. comply” evolves across token positions, identify which layers and attention heads carry the safety signal at early versus late positions, and test whether restoring early-position safety activations at later positions can causally re-trigger refusal. This approach directly connects the behavioral characterization of shallow alignment [6] with the mechanistic identification of safety components [3, 1, 9], providing the first causal account of *why* safety alignment decays during generation.

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